

Thermodynamic black box

Y21 (2021) — English translation

The thermally insulated system consists of two vessels connected by a thin heat-conducting tube. The intrinsic heat capacity of the tube can be neglected; thermal equilibrium in each vessel and along the tube is established much faster than thermal equilibrium between the vessels.

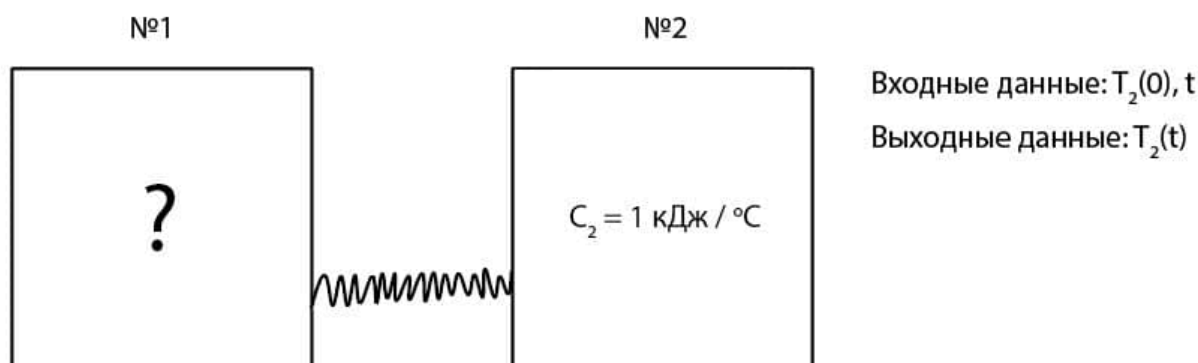


Fig. 1.

The heat capacity of the body in vessel No. 2 is known: $C_2 = 1 \text{ кДж} / ^\circ\text{C}$. You can set the initial temperature of vessel No. 2 (T_{20} , between 0°C and 100°C .) and the time t after connecting the vessels with a tube. The program offered to you shows the temperature in vessel No. 2 at time t (set in minutes) after connecting the vessels with a tube ($t = 0$).

A1	Determine the initial temperature of vessel No. 1 (T_{10}).	0.40
A2	Plot a graph of the dependence of the heat capacity of vessel No. 1 on its temperature.	3.60
A3	How long will it take for the temperature of vessel No. 1 to increase from T_{10} to $T_{10} + 20^\circ\text{C}$ if, instead of vessel No. 2, a tube is connected to a vessel with a constant temperature 100°C ?	3.00

Program

exe The executable file for Windows is published on GitHub.

Source: <https://pho.rs/p/244>